GCSE BIOLOGY

Higher Tier – Paper 2 style

Topic 4.5: Homeostasis and response

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 250 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **236**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 236

0 1 This question is about homeostasis.

0 1 . 1 What is meant by **homeostasis**? **[2 marks]**

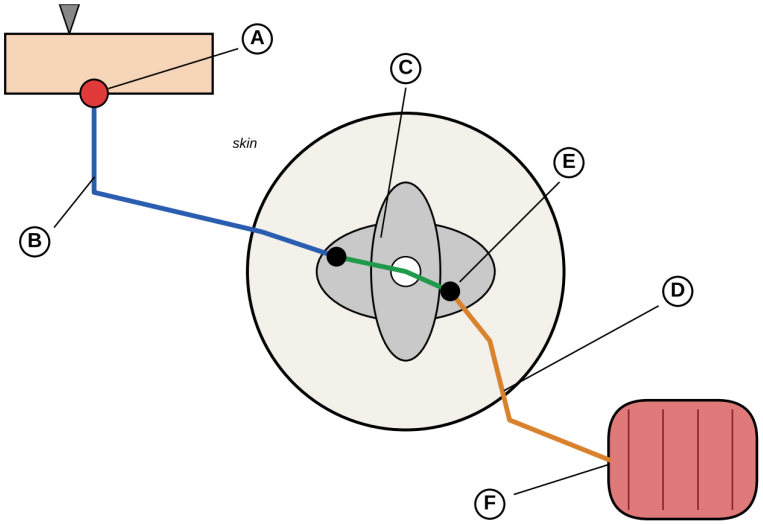
0 1 . 2 Give **three** internal conditions that are controlled in the human body. **[3 marks]**

0 1 . 3 All control systems include receptors, coordination centres and effectors. Describe the function of each. **[3 marks]**

0 1 . 4 Give **one** reason why it is important to keep body temperature constant. **[1 mark]**

0 1 . 5 Name **two** body systems that are involved in homeostasis. **[2 marks]**

0 2 A person touches a sharp pin and moves their hand away without thinking. The diagram shows the pathway of the reflex action. The letters **A** to **F** show different parts of the pathway.



0 2 . 1 Name the parts labelled **A** to **F**. **[6 marks]**

0 2 . 2 Describe the pathway taken by the nerve impulse after the pin is touched. **[3 marks]**

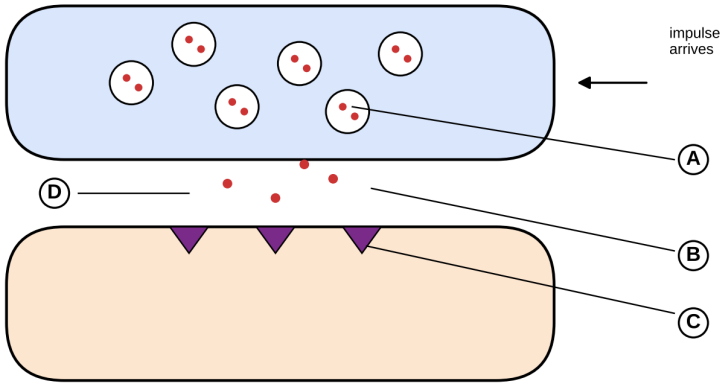
0 2 . 3 Give **two** reasons why reflex actions are important. **[2 marks]**

0 2 . 4 Name the **stimulus** and the **response** in this reflex. **[2 marks]**

0 2 . 5 Describe how an impulse passes across a synapse. **[4 marks]**

0 2 . 6 Explain why the nerve impulse can only travel in one direction across a synapse. **[2 marks]**

0 3 The diagram shows a synapse between two neurones.



0 3 . 1 Name the structures labelled **A**, **C** and **D**. **[3 marks]**

0 3 . 2 Name the chemicals labelled **B**. **[1 mark]**

0 3 . 3 Explain how a drug that blocks the receptors would affect the passing of impulses. **[2 marks]**

0 3 . 4 Explain why the response in a reflex arc is slower if there are many synapses. **[1 mark]**

0 4 A student investigates the effect of caffeine on reaction time. A partner holds a 30 cm ruler vertically. The student catches the ruler as quickly as possible after it is dropped. The distance the ruler falls before it is caught is measured. The student does the test 5 times before drinking a caffeine drink and 5 times 20 minutes after. The tables show the results and how the distance relates to reaction time.

0 4 . 1 Calculate the mean distance for the results **before** caffeine. **[2 marks]**

Test	**1**	**2**	**3**	**4**	**5**
Distance before caffeine in cm	24	19	21	18	23
Distance after caffeine in cm	17	16	18	15	19

Answer = _____ cm

0 4 . 2 The mean distance **after** caffeine is 17 cm. Use the table below to estimate the reaction time for each mean distance. **[2 marks]**

Distance in cm	**15**	**20**	**25**	**30**
Reaction time in s	0.17	0.20	0.23	0.25

0 4 . 3 What conclusion can be made from the results about the effect of caffeine on reaction time? **[1 mark]**

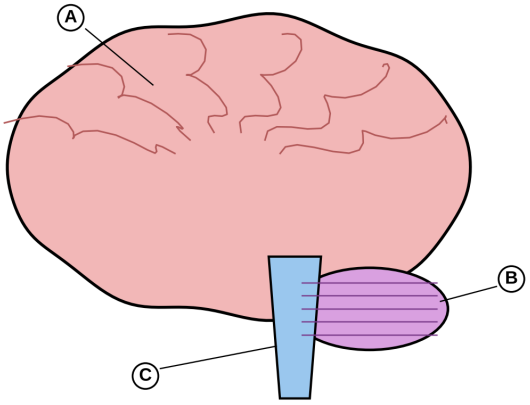
0 4 . 4 Give **two** variables the student should control. **[2 marks]**

0 4 . 5 The student's partner dropped the ruler at different times. Explain how this could affect the results. **[2 marks]**

0 4 . 6 Suggest why the student should repeat the test 5 times. **[1 mark]**

0 4 . 7 Another student's results get better as the student repeats the test many times. Suggest an explanation. **[1 mark]**

0 5 The diagram shows the human brain. The letters **A** to **C** show different regions.



0 5 . 1 Name regions **A**, **B** and **C**. **[3 marks]**

0 5 . 2 Draw one line from each region to its function. **[3 marks]**

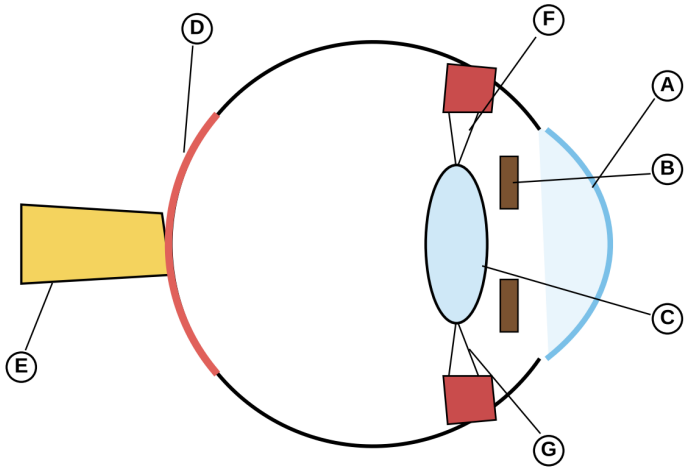
Region	Function
cerebral cortex	Controls breathing and heart rate
cerebellum	Memory, language and intelligence
medulla	Muscle coordination and balance

0 5 . 3 A patient has damage to region **B** after an accident. Predict one symptom and explain your answer. **[2 marks]**

0 5 . 4 Scientists use **three** methods to study the brain. Name **two** of them. **[2 marks]**

0 5 . 5 Suggest why it is difficult to treat brain disorders. **[2 marks]**

0 6 The diagram shows a section through the human eye. The letters **A** to **G** show different parts of the eye.

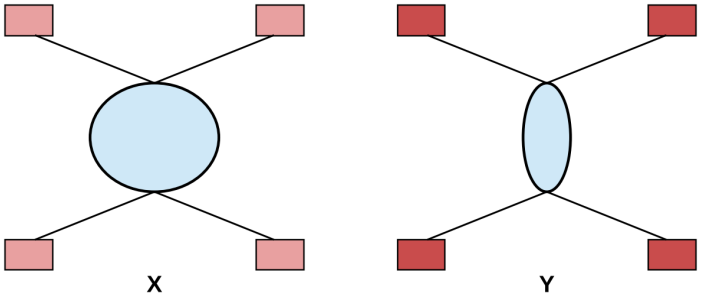


0 6 . 1 Name the parts labelled **A** to **G**. **[7 marks]**

0 6 . 2 Describe the function of the **retina**. **[2 marks]**

0 6 . 3 Describe how the eye responds when a person walks from a dim room into bright sunlight. **[3 marks]**

0 6 . 4 The diagrams **X** and **Y** show the lens when the eye is focusing on two different objects. Which diagram shows the eye focusing on a **near** object? **[1 mark]**

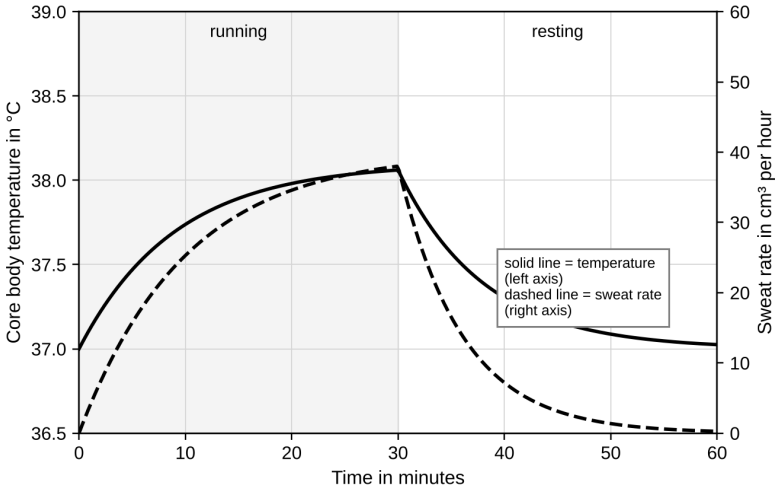


0 6 . 5 Explain how the eye changes from **Y** to **X**. **[4 marks]**

0 6 . 6 A person with **short sight (myopia)** cannot focus on distant objects. Explain what causes this and how it can be corrected. **[3 marks]**

0 6 . 7 Give **one** advantage and **one** disadvantage of laser eye surgery compared with wearing glasses. **[2 marks]**

0 7 A person runs for 30 minutes in a warm room and then rests for 30 minutes. The graph shows how the core body temperature and the sweat rate change.



0 7 . 1 State the person's core body temperature after 30 minutes. **[1 mark]**

0 7 . 2 Describe the relationship between sweat rate and core body temperature in the first 30 minutes. **[2 marks]**

0 7 . 3 Name the part of the brain that monitors the temperature of the blood. **[1 mark]**

0 7 . 4 Explain how sweating helps to lower the body temperature. **[2 marks]**

0 7 . 5 The person's blood vessels near the skin surface get wider when the body is too hot. Name this process and explain how it helps. **[3 marks]**

0 7 . 6 Describe **three** changes that happen when the body temperature falls too low. **[3 marks]**

0 7 . 7 Suggest why the temperature continues to rise for a short time after the person starts to rest. **[1 mark]**

0 7 . 8 The person sweats 40 cm^3 of water per hour at 30 minutes. Calculate the volume of water lost in 3 hours if this rate continues. **[2 marks]**

Answer = _____ cm^3

0 8 Hormones are chemicals made in glands.

0 8 . 1 Complete the table. **[6 marks]**

Gland	Hormone produced	Effect of the hormone
	FSH and LH	stimulate the ovaries
		controls metabolic rate / growth
pancreas		lowers blood glucose
adrenal gland		prepares body for fight or flight
testes		controls male secondary sexual characteristics

0 8 . 2 The pituitary gland is called the 'master gland'. Explain why. **[2 marks]**

0 8 . 3 Name the system that transports hormones around the body. **[1 mark]**

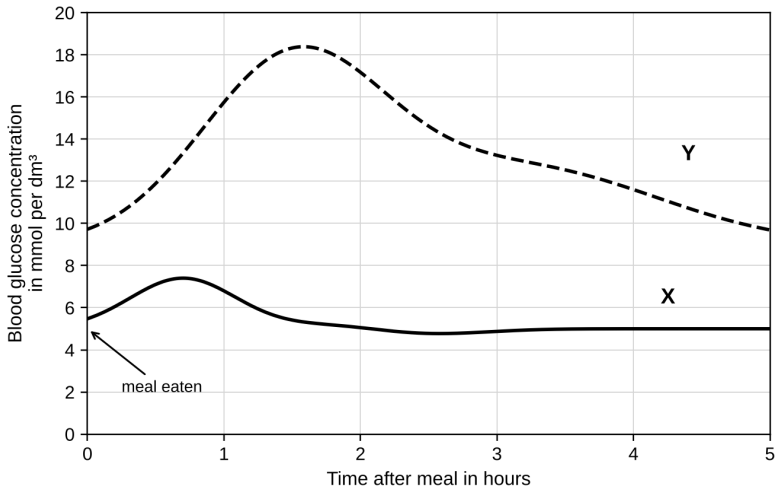
0 8 . 4 Compare the nervous system and the hormonal system. Complete the table with **two** differences. **[4 marks]**

	Nervous system	Hormonal system
Speed of response		
Duration of effect		

0 8 . 5 Describe **three** changes in the body caused by adrenaline. **[3 marks]**

0 8 . 6 Explain how thyroxine is controlled by negative feedback. **[3 marks]**

0 9 The graph shows the blood glucose concentration of two people, **X** and **Y**, after they each eat a meal containing carbohydrate. One person has type 1 diabetes.



0 9 . 1 Which person has diabetes? Give a reason for your answer. **[2 marks]**

0 9 . 2 Use the graph to calculate the difference between the highest blood glucose of **X** and the highest blood glucose of **Y**. **[2 marks]**

Answer = _____ mmol/dm³

0 9 . 3 Describe how the blood glucose concentration of person **X** returns to normal after the meal. **[3 marks]**

0 9 . 4 Explain why the blood glucose of person **Y** stays high. **[2 marks]**

0 9 . 5 Describe how type 1 diabetes is treated. **[3 marks]**

0 9 . 6 Type 2 diabetes is different from type 1. Describe the cause of type 2 diabetes. **[2 marks]**

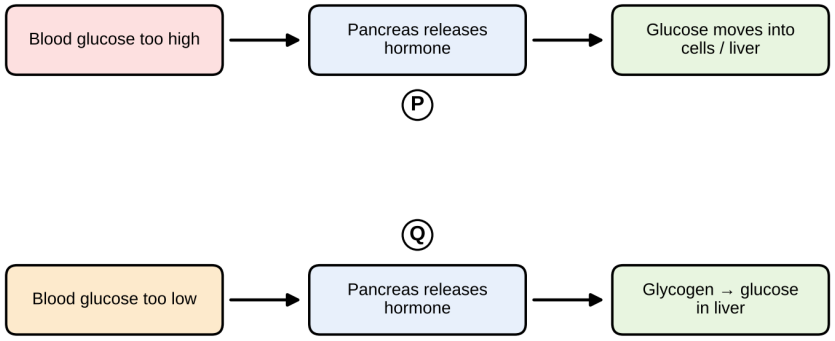
0 9 . 7 A man has a mass of 95 kg and a height of 1.70 m. Calculate his body mass index (BMI). **[3 marks]**

BMI = mass in kg ÷ (height in m)²

Answer = _____

0 9 . 8 A BMI above 30 is classed as obese. Explain why this man is more at risk of type 2 diabetes than a man with a BMI of 22. **[1 mark]**

1 0 The diagram shows how blood glucose concentration is controlled by negative feedback.



1 0 . 1 Name the hormones **P** and **Q**. **[2 marks]**

1 0 . 2 Name the organ that stores glycogen and releases glucose into the blood. **[1 mark]**

1 0 . 3 What is meant by **negative feedback**? **[2 marks]**

1 0 . 4 Explain why a person needs a different amount of insulin before and after a meal. **[2 marks]**

10.5 Suggest why athletes with type 1 diabetes may need to eat extra carbohydrate during a long race. **[2 marks]**

11 The kidneys help to control the water content and the waste substances in the blood.

11.1 Name **two** substances that are filtered out of the blood in the kidney and are **not** all reabsorbed. **[2 marks]**

11.2 Name **one** substance that is filtered out of the blood but is **all** reabsorbed. **[1 mark]**

11.3 Where is urea made in the body and from what? **[2 marks]**

11.4 Antidiuretic hormone (ADH) controls the water content of the blood. Describe what happens when the water content of the blood is too low. **[4 marks]**

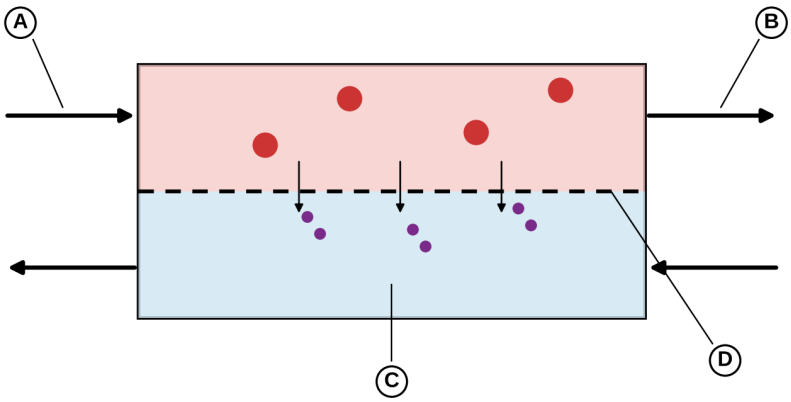
1 1 . 5 The table shows the volume of urine produced by a person in one day. Calculate the percentage decrease in the volume of urine on the hot day compared with the cool day. **[3 marks]**

Day	Urine volume in dm^3	Urine colour
cool	1.8	pale yellow
hot	0.6	dark yellow

Answer = _____ %

1 1 . 6 Explain why the urine is darker on the hot day. **[2 marks]**

1 2 A patient with kidney failure is treated by dialysis. The diagram shows how a dialysis machine works.



1 2 . 1 Name the parts of the machine labelled **C** and **D**. **[2 marks]**

12.2 Explain how urea is removed from the blood in the machine. **[3 marks]**

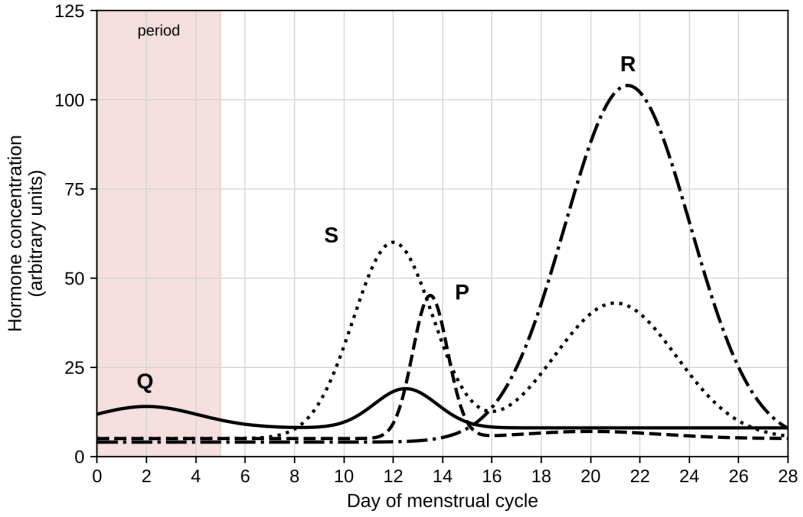
12.3 Explain why the dialysis fluid contains the same concentration of glucose and ions as normal blood. **[2 marks]**

12.4 The dialysis fluid and the blood flow in opposite directions. Suggest why. **[2 marks]**

12.5 Give **two** advantages of a kidney transplant compared with dialysis. **[2 marks]**

12.6 Give **two** problems with kidney transplants. **[2 marks]**

1 3 The graph shows the concentrations of four hormones, **P**, **Q**, **R** and **S**, during one menstrual cycle. The hormones are FSH, LH, oestrogen and progesterone.



1 3 . 1 Identify the hormones **P**, **Q**, **R** and **S**. **[4 marks]**

1 3 . 2 On which day does ovulation occur? Give a reason. **[2 marks]**

1 3 . 3 Name the organ that produces FSH and LH. **[1 mark]**

1 3 . 4 Describe the function of FSH. **[2 marks]**

1 3 . 5 Describe the function of the hormone **R**. **[2 marks]**

1 3 . 6 Explain what happens to the uterus lining after day 25 if the woman is not pregnant. **[3 marks]**

1 3 . 7 Explain how oestrogen controls the release of LH and FSH. **[3 marks]**

1 4 This question is about controlling fertility.

1 4 . 1 The table shows the number of pregnancies in one year for every 100 women using each method of contraception. **[2 marks]**

Calculate the number of pregnancies expected in one year in 2500 women who use **condoms**.

Method	**Pregnancies per 100 women per year**
No contraception	85
Condoms	18
Combined pill	8
Intrauterine device (IUD)	1
Implant	0.05

Answer = _____

1 4 . 2 Which method is the most effective? Suggest **one** reason why some women may not choose this method. **[2 marks]**

1 4 . 3 Explain how the combined pill prevents pregnancy. **[3 marks]**

1 4 . 4 Give **one** advantage of using a condom over using the pill. **[1 mark]**

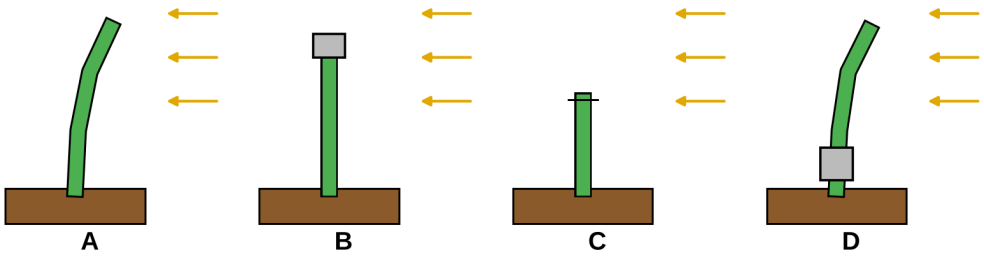
1 4 . 5 Describe how IVF can help a woman who cannot become pregnant naturally. **[4 marks]**

1 4 . 6 In one year a clinic carried out 5600 IVF treatments and 1400 resulted in a baby. Calculate the success rate. **[2 marks]**

Answer = _____ %

14.7 Give **two** disadvantages of IVF. **[2 marks]**

15 A student investigates how shoots respond to light. The diagram shows four seedlings **A** to **D** after 3 days. Each was lit from one side. Seedling **B** had a foil cap on the tip. Seedling **C** had its tip removed. Seedling **D** had foil around the base.



15.1 What is the name for the growth response of a shoot towards light? **[1 mark]**

15.2 Name the plant hormone that causes cell elongation in the shoot. **[1 mark]**

15.3 Explain why seedling **A** bends towards the light. **[3 marks]**

15.4 Explain the results for seedlings **B** and **C**. **[3 marks]**

1 5 . 5 Explain why seedling **D** still bends. **[1 mark]**

1 5 . 6 Describe the role of gibberellins and ethene in plants. **[2 marks]**

1 5 . 7 Give **two** uses of plant hormones in agriculture and horticulture. **[2 marks]**

1 5 . 8 A shoot grows upwards and a root grows downwards. Name these responses to gravity. **[2 marks]**

1 6 Stretch questions (grade 7–9).

1 6 . 1 *Describe how blood glucose is controlled in a healthy person and explain how the control differs in a person with type 1 diabetes. **[6 marks]**

1 6 . 2 *Explain how the hormones FSH, LH, oestrogen and progesterone interact to control the menstrual cycle. **[6 marks]**

1 6 . 3 *A person spends a day running in the desert without drinking water. Explain the changes that occur in their body to control body temperature and water content. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.5: Homeostasis and response

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	the regulation of the internal conditions of a cell or organism ; to maintain optimum conditions for function / enzyme action in response to internal and external changes ;		2	AO1 4.5.1
01.2	body temperature ; blood glucose concentration ; water content / ion content / pH ;	Any three.	3	AO1 4.5.1
01.3	receptors detect stimuli (changes in the environment) ; coordination centres (brain, spinal cord, pancreas) receive and process information ; effectors (muscles or glands) bring about responses that restore optimum levels ;		3	AO1 4.5.1
01.4	enzymes have an optimum temperature ; so that metabolic reactions work at the best rate (not denatured) ;	Any one.	1	AO2 4.5.1
01.5	nervous system ; endocrine (hormonal) system ;		2	AO1 4.5.1
02.1	A: receptor ; B: sensory neurone ; C: relay neurone ; D: motor neurone ; E: synapse ; F: effector (muscle) ;	One mark each. Allow 'synapse' for E only if it is the junction shown.	6	AO1 4.5.2.1
02.2	receptor → sensory neurone → relay neurone (in the spinal cord / CNS) ; → motor neurone ; → effector (muscle contracts) ;		3	AO1 4.5.2.1
02.3	they are automatic / do not involve thinking ; they are rapid, so they protect the body from harm ;		2	AO1 4.5.2.1
02.4	stimulus: the pin / pain / sharp object ; response: muscle contracts to move the hand away ;		2	AO2 4.5.2.1
02.5	the impulse arrives at the end of one neurone ; chemicals (neurotransmitter) are released into the gap ; they diffuse across the gap ; and bind to receptors on the next neurone which starts a new impulse ;	Triple only (the detail of the chemical transmitter).	4	AO1 4.5.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	the neurotransmitter is only released from the first neurone ; receptors are only on the second neurone ;		2	AO2 4.5.2.1
03.1	A: vesicle (containing neurotransmitter) ; C: receptor (molecule) ; D: (synaptic) gap ;		3	AO1 4.5.2.1
03.2	neurotransmitter(s)		1	AO1 4.5.2.1
03.3	the neurotransmitter cannot bind ; so the impulse is not passed on / no response ;		2	AO3 4.5.2.1
03.4	diffusion across each gap takes time ;		1	AO3 4.5.2.1
04.1	(24 + 19 + 21 + 18 + 23) ÷ 5 ; 21 cm ;		2	AO2 4.5.2.2 / RP
04.2	before: 0.21 s (allow 0.20–0.22) ; after: 0.19 s (allow 0.18–0.20) ;		2	AO3 4.5.2.2 / RP
04.3	caffeine reduces / shortens the reaction time ;		1	AO3 4.5.2.2 / RP
04.4	same hand / same ruler / same person dropping ; time of day / same amount of caffeine / no distractions ;		2	AO3 WS 2.3
04.5	student could predict when the ruler will fall ; which reduces the reaction time (results less valid) ;		2	AO3 4.5.2.2 / RP
04.6	so a mean can be calculated / reduce the effect of random errors ;		1	AO3 WS 3.4
04.7	practice improves the pathway / the brain learns the response / predicting ;		1	AO3 4.5.2.2
05.1	A: cerebral cortex ; B: cerebellum ; C: medulla (oblongata) ;		3	AO1 4.5.2.2
05.2	cerebral cortex — consciousness, intelligence, memory and language ; cerebellum — coordinates movement and balance ; medulla — controls unconscious activities such as heart rate and breathing ;		3	AO1 4.5.2.2
05.3	clumsy movement / poor balance / jerky movements ; because cerebellum coordinates muscle activity / balance ;		2	AO3 4.5.2.2
05.4	MRI scans ; studying patients with brain damage / electrically stimulating the brain ;	Allow CT scans. Any two.	2	AO1 4.5.2.2
05.5	the brain is complex / delicate / hard to access ; so risk of damage in surgery / drugs cannot reach / many unknowns ;		2	AO3 4.5.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	A: cornea ; B: iris ; C: lens ; D: retina ; E: optic nerve ; F: ciliary muscle ; G: suspensory ligaments ;	One mark each.	7	AO1 4.5.2.3
06.2	contains receptor cells / rods and cones ; sensitive to light and colour / sends impulses along the optic nerve ;		2	AO1 4.5.2.3
06.3	circular muscles of the iris contract ; radial muscles relax ; the pupil gets smaller so less light enters (protects the retina) ;		3	AO2 4.5.2.3
06.4	X ;		1	AO2 4.5.2.3
06.5	the ciliary muscles contract ; the suspensory ligaments loosen ; the lens becomes thicker / more curved ; so light is refracted (bent) more to focus on the retina ;		4	AO2 4.5.2.3
06.6	the eyeball is too long / the lens is too curved so light focuses in front of the retina ; corrected by wearing a concave (diverging) lens ; alternatives: laser surgery / contact lenses ;		3	AO2 4.5.2.3
06.7	advantage: permanent correction / no glasses ; disadvantage: risk of infection / may damage the eye / cost ;		2	AO3 4.5.2.3
07.1	38.1 (°C) (allow 38.0–38.2)		1	AO3 WS 3.1
07.2	both increase ; sweat rate rises as the temperature rises ;		2	AO3 4.5.2.4
07.3	hypothalamus / thermoregulatory centre		1	AO1 4.5.2.4
07.4	sweat evaporates from the skin ; taking energy / heat from the body ;		2	AO2 4.5.2.4
07.5	vasodilation ; more blood flows near the surface of the skin ; so more heat is transferred to the surroundings by radiation ;		3	AO2 4.5.2.4
07.6	vasoconstriction (blood vessels narrow) ; shivering (muscles contract releasing heat from respiration) ; sweating stops / hairs stand on end ;		3	AO1 4.5.2.4
07.7	muscles still releasing heat / heat has not yet been lost ;	Allow: delay in response.	1	AO3 4.5.2.4
07.8	40 × 3 ; 120 cm^3 ;		2	AO2 MS 1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	pituitary ; thyroid ; thyroxine ; insulin ; adrenaline ; testosterone ;	One mark for each blank cell. Allow ADH / growth hormone for pituitary.	6	AO1 4.5.3.1
08.2	it produces hormones that act on other glands ; causing them to release hormones / controls other glands ;		2	AO2 4.5.3.1
08.3	blood / circulatory system		1	AO1 4.5.3.1
08.4	nervous: fast response ; hormonal: slower response ; nervous: short-lasting effect ; hormonal: long-lasting effect ;	Allow nervous: electrical impulses / hormonal: chemicals in the blood.	4	AO2 4.5.3.1
08.5	heart rate increases ; more blood flows to the muscles ; more glucose released / more oxygen to muscles / pupils dilate ;		3	AO1 4.5.3.1
08.6	thyroxine levels fall → the pituitary releases TSH ; TSH stimulates the thyroid to release thyroxine ; when levels rise the pituitary releases less TSH ;	Triple only. Allow hypothalamus / TRH.	3	AO2 4.5.3.2
09.1	Y ; blood glucose is much higher / does not return to normal ;		2	AO3 4.5.3.2
09.2	Y = 18.4 and X = 7.4 (allow ±0.4) ; 11 mmol per dm^3 (allow 10.5–11.5) ;		2	AO2 WS 3.1
09.3	blood glucose rises and the pancreas detects this ; the pancreas releases insulin ; glucose moves into cells / liver converts glucose to glycogen ;		3	AO2 4.5.3.2
09.4	the pancreas does not produce enough / any insulin ; so glucose cannot be taken up / converted to glycogen ;		2	AO2 4.5.3.2
09.5	injections of insulin ; monitoring blood glucose / carbohydrate in diet ; avoid sugary foods / take regular exercise ;		3	AO1 4.5.3.2
09.6	body cells no longer respond to insulin ; obesity is a risk factor / or not enough insulin produced ;		2	AO1 4.5.3.2
09.7	$height^2 = 2.89$; 95 ÷ 2.89 ; 32.9 (allow 33) ;		3	AO2 MS 1a
09.8	obesity is a risk factor for type 2 diabetes ;		1	AO3 4.5.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.1	P: insulin ; Q: glucagon ;		2	AO1 4.5.3.2
10.2	liver		1	AO1 4.5.3.2
10.3	a change in a condition causes a response ; that reverses the change / returns it to the normal level ;		2	AO1 4.5.3.2
10.4	glucose from food raises blood glucose after a meal ; more insulin is needed to lower it / less when fasting ;		2	AO3 4.5.3.2
10.5	exercise uses more glucose (respiration in muscles) ; so blood glucose may fall too low (less insulin needed) ;		2	AO3 4.5.3.2
11.1	urea ; (excess) ions / water ;		2	AO1 4.5.3.3
11.2	glucose		1	AO1 4.5.3.3
11.3	liver ; from the breakdown of excess amino acids ;		2	AO1 4.5.3.3
11.4	the pituitary gland releases more ADH ; ADH makes the kidney tubules reabsorb more water ; a small volume of concentrated urine is produced ; the water content of the blood returns to normal (negative feedback) ;		4	AO2 4.5.3.3
11.5	1.8 − 0.6 = 1.2 ; 1.2 ÷ 1.8 × 100 ; 66.7 % (67 %) ;		3	AO2 MS 1c
11.6	more water lost in sweat so more ADH released ; urine is more concentrated (less water, same urea) ;		2	AO3 4.5.3.3
12.1	C: dialysis fluid ; D: partially permeable membrane ;		2	AO1 4.5.3.3
12.2	urea diffuses ; from the blood to the dialysis fluid through the membrane ; down a concentration gradient ;		3	AO2 4.5.3.3
12.3	so that glucose / useful ions are not lost from the blood ; there is no concentration gradient for them ;		2	AO2 4.5.3.3
12.4	keeps a concentration gradient along the whole length ; so diffusion of urea happens all the way / more efficient ;		2	AO3 4.5.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.5	no need for regular dialysis sessions / more freedom ; cheaper in the long term / better quality of life ;		2	AO3 4.5.3.3
12.6	shortage of donors ; the donor kidney may be rejected / immunosuppressant drugs needed ;		2	AO1 4.5.3.3
13.1	P: LH ; Q: FSH ; R: progesterone ; S: oestrogen ;		4	AO3 4.5.3.5
13.2	about day 14 ; just after the LH surge / peak (P) ;		2	AO3 4.5.3.5
13.3	pituitary gland		1	AO1 4.5.3.5
13.4	causes eggs to mature in the ovary / follicle ; stimulates the ovaries to produce oestrogen ;		2	AO1 4.5.3.5
13.5	maintains the lining of the uterus ; (the lining is ready to receive an embryo / prevents menstruation) ;		2	AO1 4.5.3.5
13.6	progesterone level falls ; the lining breaks down ; menstruation / period begins ;		3	AO2 4.5.3.5
13.7	oestrogen levels rise as the follicle grows ; this inhibits FSH (negative feedback) ; high oestrogen stimulates the LH surge (positive feedback) ;	Higher Tier.	3	AO2 4.5.3.5
14.1	18 ÷ 100 × 2500 ; 450 ;		2	AO2 4.5.3.6 / MS 1c
14.2	implant ; needs a minor procedure / side effects / cannot be removed quickly (must be fitted by a doctor) ;		2	AO3 4.5.3.6
14.3	contains oestrogen / progesterone ; oestrogen inhibits FSH so no eggs mature ; progesterone thickens cervical mucus so sperm cannot pass ;		3	AO2 4.5.3.6
14.4	protects against sexually transmitted infections ;		1	AO3 4.5.3.6
14.5	FSH and LH injections to stimulate eggs to mature ; eggs collected from the ovaries ; fertilised by sperm in a laboratory ; embryos placed in the uterus ;		4	AO1 4.5.3.7
14.6	1400 ÷ 5600 × 100 ; 25 % ;		2	AO2 MS 1c
14.7	physically / emotionally stressful ; low success rate / may lead to multiple births / expensive ;		2	AO3 4.5.3.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
15.1	phototropism		1	AO1 4.5.4
15.2	auxin		1	AO1 4.5.4
15.3	auxin is produced in the tip and moves down ; it collects on the shaded side ; cells on the shaded side elongate faster so the shoot bends towards the light ;		3	AO2 4.5.4
15.4	B: tip covered so no light detected / auxin spread evenly so grows straight ; C: tip removed so no auxin produced ; so no (elongation) growth ;		3	AO3 4.5.4
15.5	the tip (detecting light) is not covered ;		1	AO3 4.5.4
15.6	gibberellins: end dormancy of seeds / promote germination / flowering ; ethene: controls cell division and ripening of fruits ;		2	AO1 4.5.4
15.7	auxins in rooting powders / weedkillers ; ethene to control ripening / gibberellins to produce seedless fruit ;	Any two.	2	AO1 4.5.4
15.8	shoot: negative gravitropism ; root: positive gravitropism ;		2	AO1 4.5.4
16.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • pancreas detects the concentration of glucose in the blood (coordination centre) • if too high the pancreas secretes insulin; glucose moves into cells and the liver converts it to glycogen • if too low the pancreas secretes glucagon; glycogen is converted to glucose in the liver • this is negative feedback; it returns the glucose concentration to normal • type 1: the pancreas does not make (enough) insulin so glucose stays high after a meal • treatment: insulin injections, controlled carbohydrate, monitoring; compare with type 2 (insulin resistance)		6	AO2 4.5.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • FSH from the pituitary causes an egg to mature in a follicle and stimulates oestrogen production • oestrogen causes the lining of the uterus to build up / repair • high oestrogen inhibits FSH (negative feedback) and stimulates the LH surge • LH causes ovulation at about day 14 • after ovulation the follicle forms the corpus luteum which makes progesterone; progesterone maintains the lining • if no pregnancy, progesterone falls, the lining breaks down (menstruation) and the cycle restarts		6	AO2 4.5.3.5
16.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • temperature rises: thermoreceptors / the hypothalamus detect it; sweating increases to cool the body by evaporation • vasodilation: more blood flows to the skin surface so more heat is lost • water lost in sweat reduces the water content of the blood • pituitary gland releases more ADH • kidney tubules reabsorb more water; a small volume of concentrated urine is produced • negative feedback: responses restore both temperature and water level; dehydration limits sweating if not corrected		6	AO2 4.5.2.4 / 4.5.3.3

Where marks are lost in Topic 4.5

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Homeostasis** = keeping conditions steady using receptor → coordination centre → effector. Say 'negative feedback' and explain what is reversed.
- ☐ **Reflex arc:** stimulus → receptor → sensory neurone → relay neurone → motor neurone → effector → response. Know the synapse: neurotransmitter diffuses across the gap.
- ☐ **Reaction time:** repeat and take a mean; control the person, the hand, the ruler and the distraction. A shorter distance means a faster reaction.
- ☐ **Eye:** near object = ciliary muscles contract, ligaments loosen, lens thick. Far object = muscles relax, ligaments tight, lens thin. Bright light = circular muscles contract, pupil small.
- ☐ **Temperature:** too hot = sweat + vasodilation; too cold = shiver + vasoconstriction. Name the process and link it to heat loss.
- ☐ **Blood glucose:** insulin lowers it (glucose → glycogen); glucagon raises it. Type 1 = no insulin (injections); type 2 = cells resistant (diet, exercise).
- ☐ **Kidney:** urea is made in the liver. ADH is made by the pituitary and increases water reabsorption. Dialysis = diffusion of urea through a partially permeable membrane.
- ☐ **Menstrual cycle:** FSH matures egg; oestrogen builds lining; LH triggers ovulation (day 14); progesterone maintains the lining. Learn which line is which on the graph.
- ☐ **Percentage calculations:** (change ÷ original) × 100. For 'per 100 women' data, multiply by the group size ÷ 100.
- ☐ **Plant hormones:** auxin accumulates on the shaded side → cells elongate faster → bends towards light. Tip removed = no auxin = no growth.